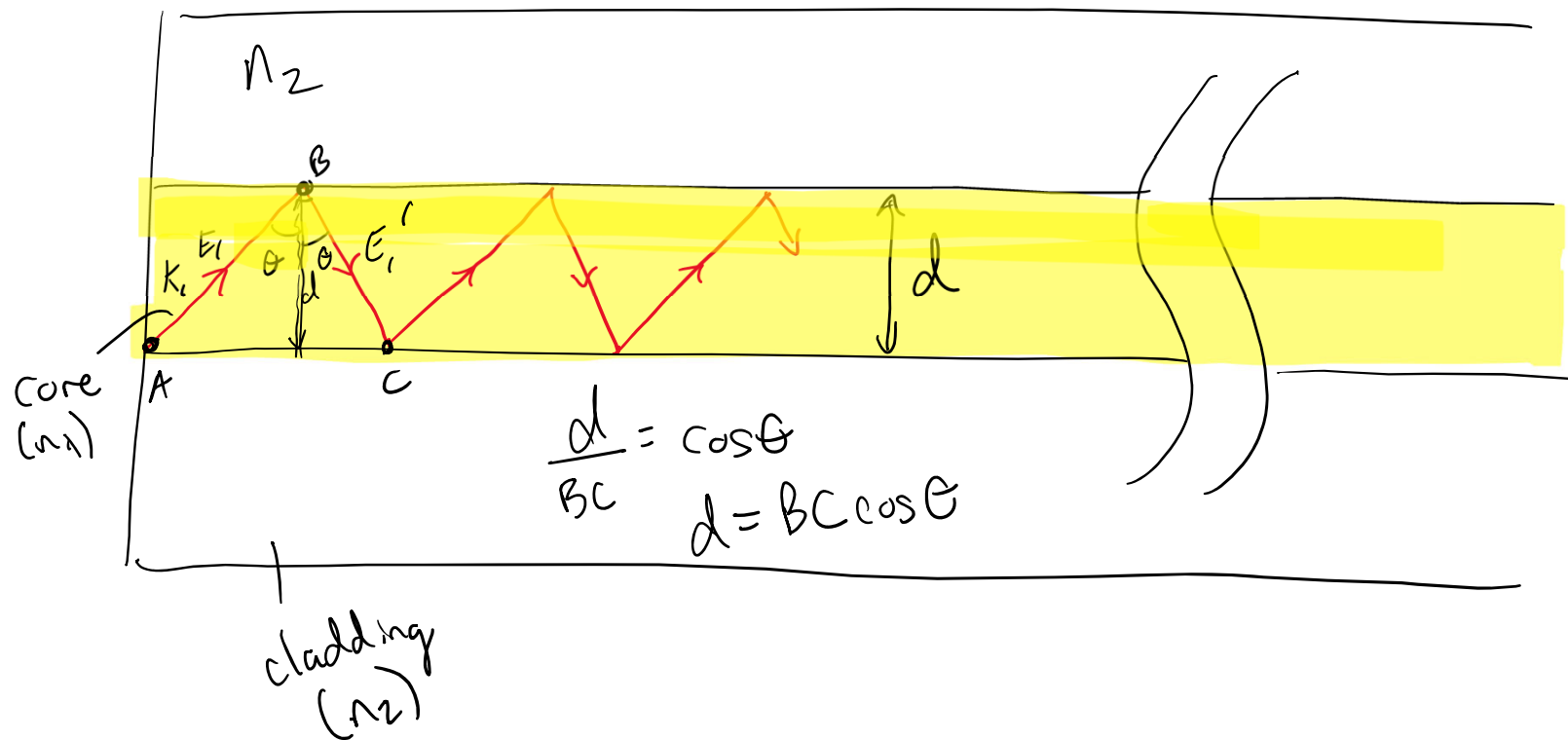


Deriving the waveguide condition



the oscillation phase (ϕ) b/w A to B, and B to C
 E_i will have a slight phase change with E_i'

Related by the path

$$k_i = k_0(n_1) = \frac{2\pi(n_1)}{\lambda_0}$$

$$\Delta \Phi_{AC} = k_i(AB + BC) - 2\phi = m(2\pi); m = 0, 1, 2, 3, \dots$$

mode (vertical direction)

$$k_i(BC \cos 2\theta + BC)$$

$$k_i BC [2\cos^2 \theta - 1] + 1$$

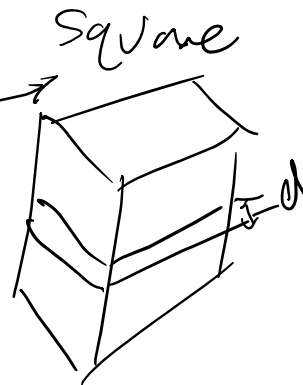
$$BC \cos 2\theta + BC = BC [2 \cos^2 \theta - 1]$$

$$= \frac{d}{\cos \theta} [2 \cos^2 \theta]$$

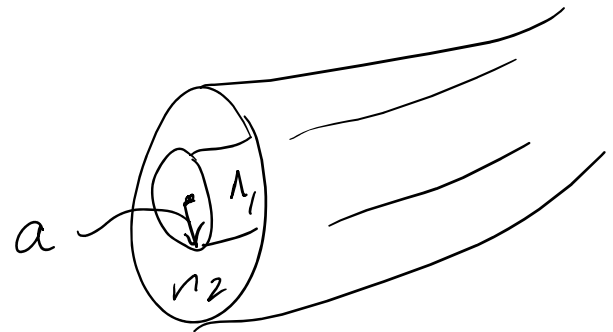
$$BC \cos 2\theta + BC = 2d \cos \theta$$

$$k_1 (2d \cos \theta) - \phi = m(2\pi)$$

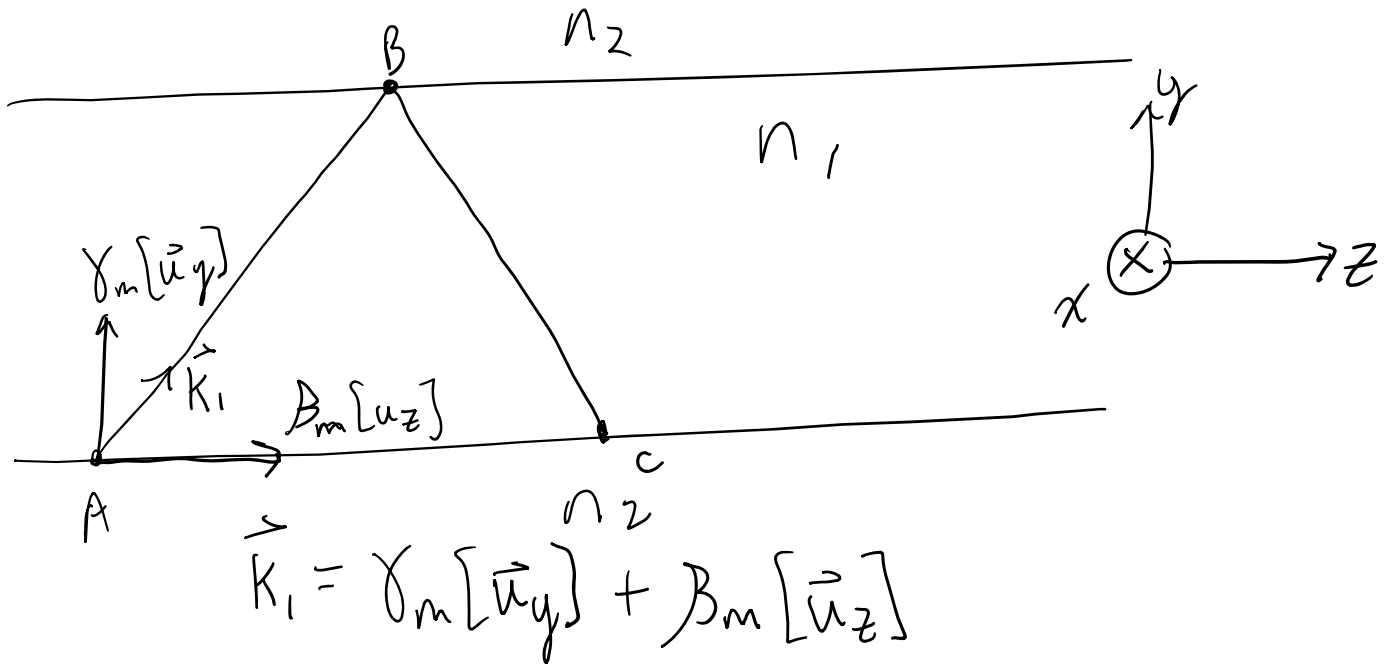
$$\left(\frac{2\pi n_1 d}{\lambda_0} \right) \cos \theta - \phi = m\pi$$



$$\frac{2\pi n_1 (2a)}{\lambda_0} \cos \theta - \phi = m\pi$$



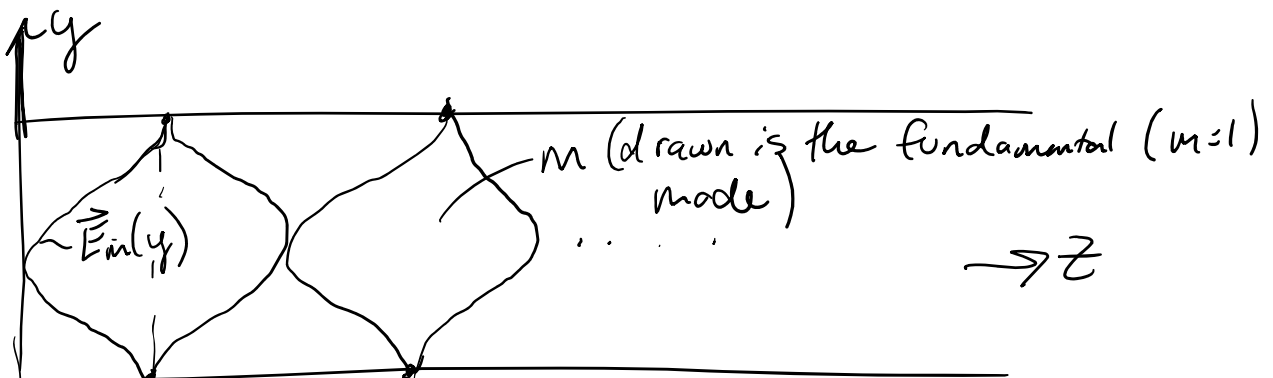
k_1 (propagation constant) can be split into two components



$$|\beta_m| = |k_1| \sin \theta = \frac{2\pi n_1}{\lambda_0} \sin \theta$$

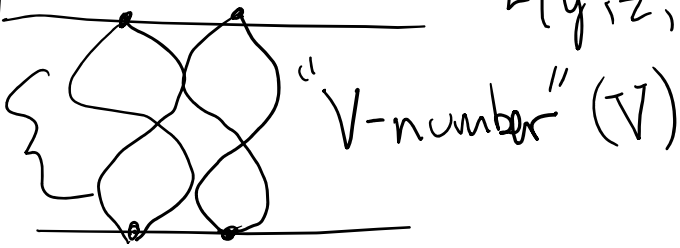
$\lambda_0 \leftarrow$ freespace wavelength

$$\gamma_m = |k_1| \cos \theta = \frac{2\pi n_1}{\lambda_0} \cos \theta$$



$$\vec{E}(y, z, t) = E_m(y) \cos(\omega t - (\beta_m z) + \psi)$$

mode numbers



ψ is any phase at time $t=0$

V-number: is a dimensionless parameter that determines the number of modes in a fiber

$$m \leq \frac{2V - \phi}{\pi} \quad \leftarrow \begin{array}{l} \text{interference phase from one reflection} \\ \text{to another} \end{array}$$

also called
"V-parameter"

* the V-number or V-parameter is normalized in freq.

• sometimes (depending on manufacturer) it is normalized in core thickness.

If we assume a circular fiber

$$V = \frac{2\pi a}{\lambda_0} (n_1^2 - n_2^2)^{1/2}$$

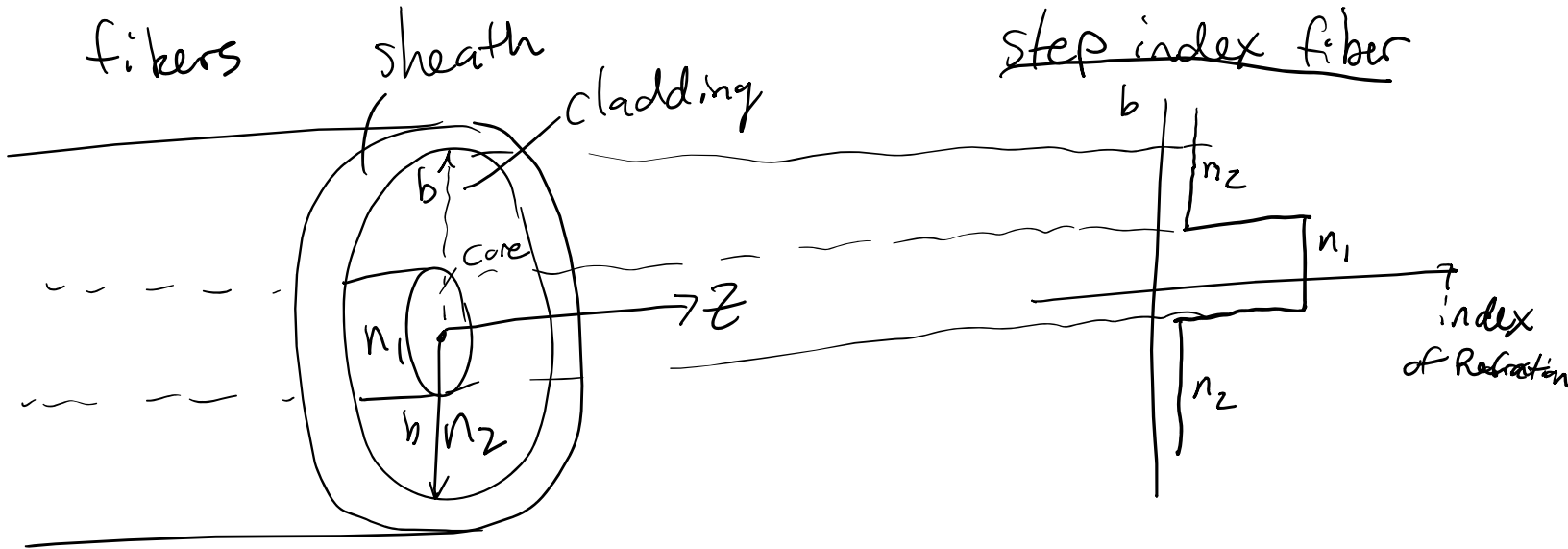
due to evanescence at the n_1, n_2 boundary there will be a penetration of energy into the cladding

Penetration Depth into cladding

$$\frac{1}{\delta_m} = \alpha_m = \frac{2\pi n_2}{\lambda_0} \left[\left(\frac{n_1}{n_2} \right)^2 \sin^2 \theta - 1 \right]^{1/2}$$

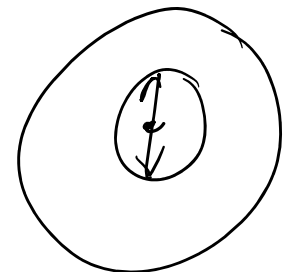
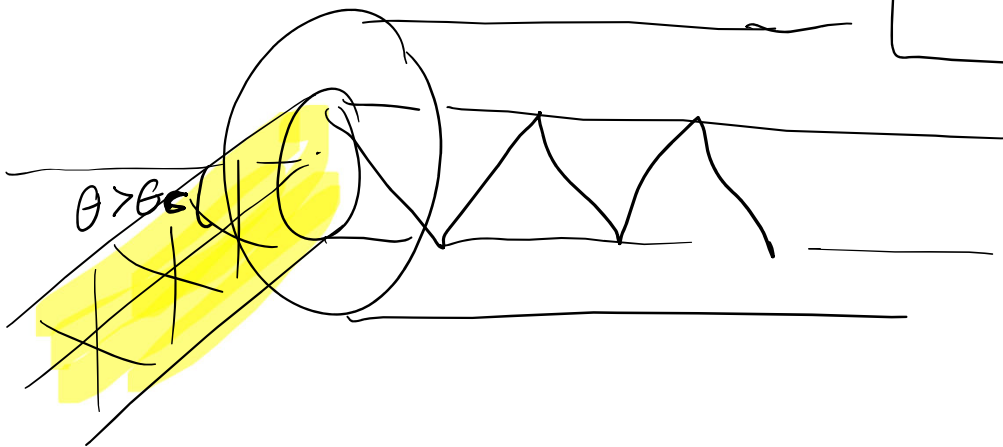
How we arrange the indexes of Refraction in fibers

step index fiber

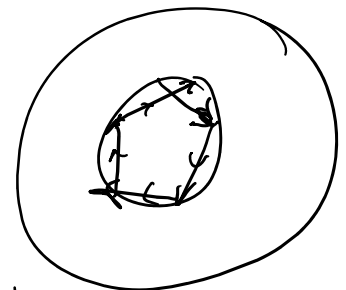
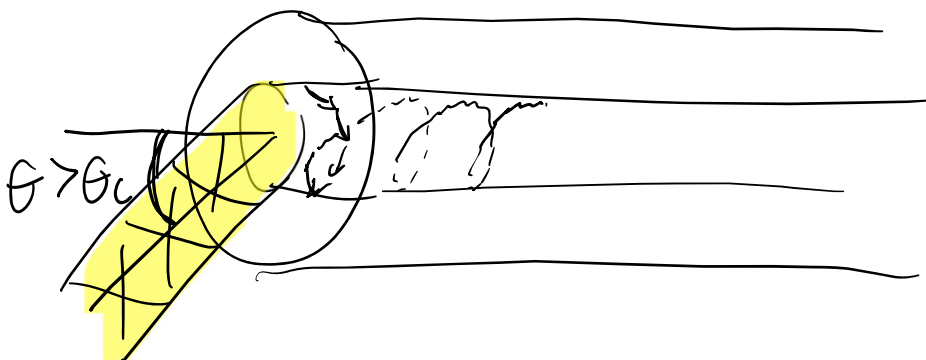


*normalize the indexes (Δ)

$$\Delta = \frac{n_1 - n_2}{n_1}$$



meridional Ray
m-interface



Skew Ray
l-interface

* both are present in fiber.